Executive summary of the report

This is a descriptive analysis of share market trader investment portfolio dataset which is further on used for forecasting the next five trading days; it consists of 179 observations taking from out of 252 possible trading days. We have converted the dataset into a time series data then applied model building strategies by model identification, model fitting and model diagnosis using various functions on R. After finding the best fit model for our series we have done forecasting using that model.

Table of contents

Introduction

In this report, we have done analysis leading to forecasting wherein we are using a dataset which consists of returns of a share market trader's investment portfolio. The data was collected on consecutive days, except the weekends, out of 252 trading days in a year. Using this data after applying all the statistical tools, we do the forecasting for next five trading days.

Methodology

1. R programming software used for analysis
2. Data interpretation and conversion
3. Correlation between consecutive lags
4. Model estimation
5. Forecasting with the best fit model

R programming software

All the data was analysed using R, library package TSA has been used for time series data and for performing various other descriptive analysis on the data.

Data interpretation and conversion

First, we cleared the R environment, then loaded the TSA library package.

```
# To clean all the R memory
rm(list=ls())

# Loading time series package in R
library(TSA)
```

Figure 1

Setting up our working directory.

```
# setting up the working directory
setwd("C:/Users/shamb/OneDrive/Desktop/Time Series Analysis/Assignment 1")
```

Figure 2

Reading our dataset in R

```
# Reading the data file of Share Market Trader's Investment Portfolio
ShareMarket <- read.csv("assignment1Data2024.csv")
```

Figure 3

View the dataset

```
# View the loaded file
ShareMarket
head(ShareMarket)
class(ShareMarket)
plot(ShareMarket, type='o', ylab= 'Share Market Investment')
```

Figure 4

Output

```
> head(ShareMarket)
  X        x
1 1 149.9764
2 2 147.3312
3 3 143.7421
4 4 141.4270
5 5 140.1138
6 6 139.0057
> class(ShareMarket)
[1] "data.frame"
```

Figure 5

Output

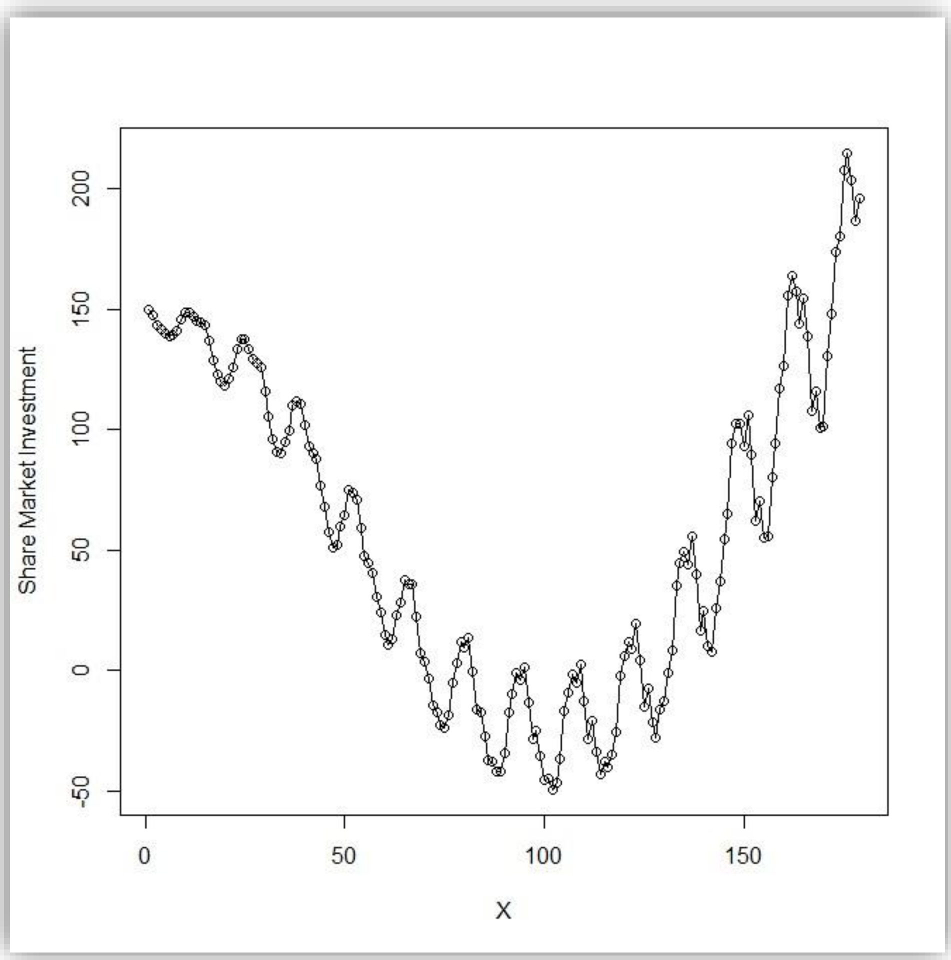

Figure 6

Converting dataset into a time series data

```
# Convert dataset ShareMarket to a time series data
ShareMarketTS <- ts((ShareMarket$x), frequency = 5)
ShareMarketTS
```

Figure 7

Viewing the share market time series data

```
# View the TS data file
class(ShareMarketTS)
plot(ShareMarketTS, type='o', ylab= 'Investment return amount', main= "Time series plot of share market trader investment")
summary(ShareMarketTS)
```

Figure 8

Output

```
> class(ShareMarketTS)
[1] "ts"
> plot(ShareMarketTS, type='o', ylab= 'Investment
> summary(ShareMarketTS)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-49.167  -2.685  51.105  57.043 117.781 214.611
```

Figure 9

Output

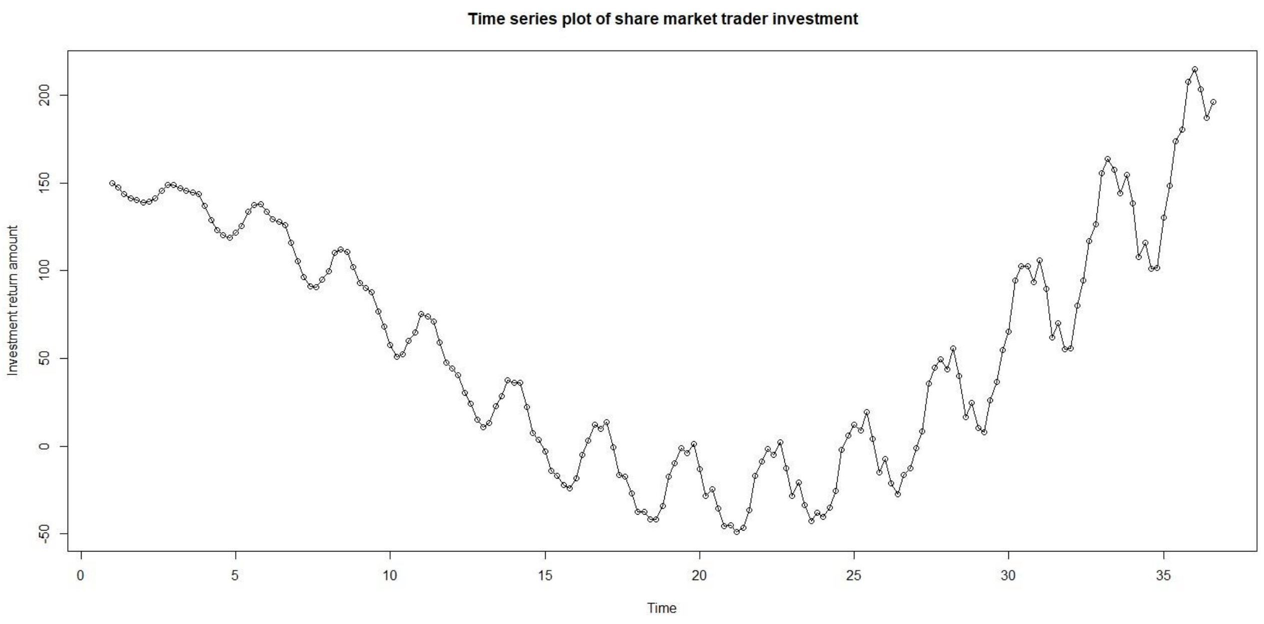

Figure 10

Analysing five points of interest:

Trend: From the given plot of the time series, we can see a significant upward trend which is also slightly quadratic.

Changing variance: There is no such changing variance in the time series as per shown in the graph.

Seasonality: There is some seasonality in the series, as there is a slight repeating pattern during the start of the plot and then again towards to the end.

Behaviour pattern: Behaviour of the time series appears to be autoregressive i.e. AR, since we can see the successive data points.

Change point: There is no significant change point in the series, but we can say that somewhere around point 20 on x-axis the graph starts going upwards. So, there might be some change happening around there in the time series.

Correlation between consecutive lags

We checked for two lags in our series. Lags are created so that we can see if there is any correlation between consecutive days.

First lag:

```
# Creating first lag
orig = ShareMarketTS
l = zlag(ShareMarketTS)
index = 2:length(orig)
cor(orig[index],l[index])
```

Figure 11

Output

```
> orig = ShareMarketTS
> l = zlag(ShareMarketTS)
> index = 2:length(orig)
> cor(orig[index],l[index])
[1] 0.9868369
>
```

Figure 12

Inference: as we have got the correlation between t and t-1 day to be 0.9868369, correlation value has come to be large signifying a positive strong correlation.

Second lag:

```
# Creating second lag
l2 = zlag(zlag(ShareMarketTS))
index = 3:length(l2)
cor(orig[index],l2[index])
```

Figure 13

Output

```
> l2 = zlag(zlag(ShareMarketTS))
> index = 3:length(l2)
> cor(orig[index],l2[index])
[1] 0.963663
```

Figure 14

Inference: correlation between t and t-2 day has come out to be equally large, again signifying a positive strong correlation.

ACF plot of lags:

```
# ACF plot of lags
acf(ShareMarketTS, lag.max = 50)
```

Figure 15

Output

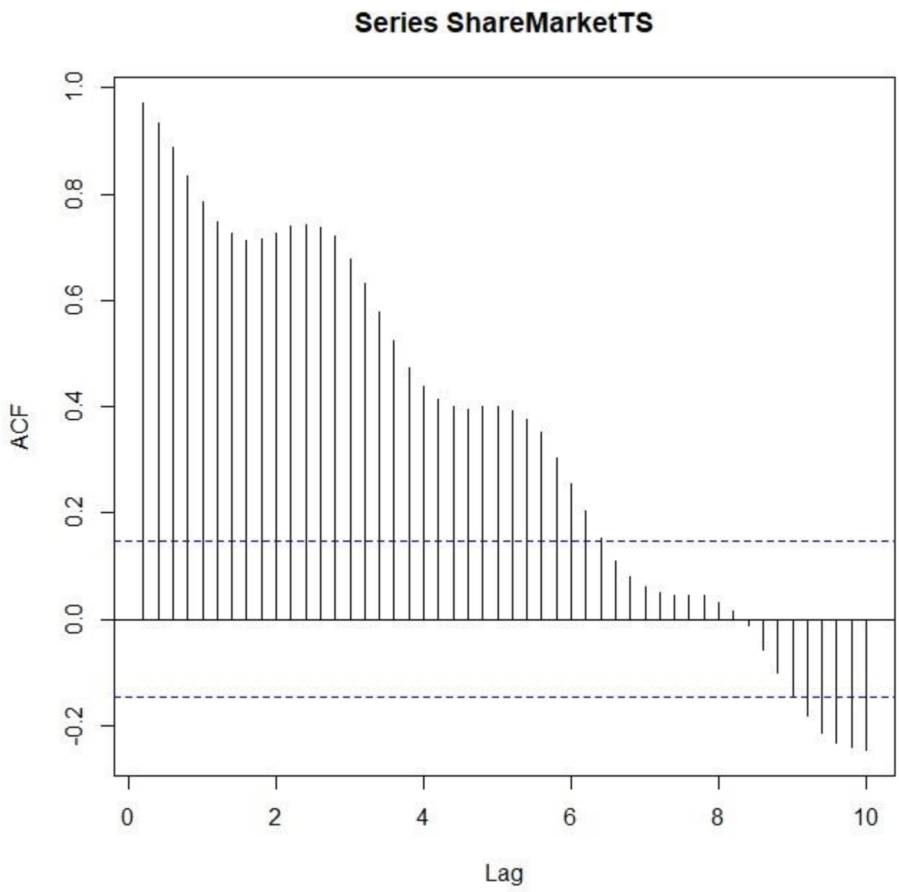

Figure 16

Inference: from the ACF plot of lags, we can see that almost all the lags are significant which shows that the series has an autoregressive behaviour. However, the wave pattern in lags suggests that there is a presence of seasonality in the time series data.

Model Estimation

For model estimation we have used regression analysis, we use regression analysis for non-stationary/ non-constant series. The following models have been used: linear model, quadratic model, seasonal model, cosine model, quadratic plus seasonal model. After trying these models to fit in our series we then do model diagnosis and model interpretation.

1. Linear model

Fitting the linear model

```
# Fitting the linear model
t <- time(ShareMarketTS)
dataFreq <- frequency(ShareMarketTS)
linearmodel = lm(ShareMarketTS ~ t)
summary(linearmodel)
```

Figure 17

Output

```
Call:
lm(formula = ShareMarketTS ~ t)

Residuals:
     Min       1Q   Median       3Q      Max
-104.186  -58.689   -5.424   57.532  172.072

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  72.8955    10.5492   6.910 8.42e-11 ***
t            -0.8432     0.4917  -1.715   0.0881 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 67.99 on 177 degrees of freedom
Multiple R-squared:  0.01634,   Adjusted R-squared:  0.01079
F-statistic: 2.941 on 1 and 177 DF,  p-value: 0.08812
```

Figure 18

Inference: the coefficient of the interpret has come out to be 72.8955 and coefficient of independent variable t is -0.8432, so the intercept value is statistically significant however the coefficient of independent variable is statistically insignificant at 5% level. F-statistics has a p-value of 0.08812 which is greater than 0.05 so it statistically insignificant. Adjusted R-squared is 0.01079 which is low suggesting that the linear model wouldn't be a good fit for the series.

```
fitted.linearmodel<- fitted(linearmodel)
plot(ShareMarketTS,type='o',ylab='Return_Amount', xlab='Day', ylim = c(-49.167,214.611))
abline(linearmodel)
```

Figure 19

Output

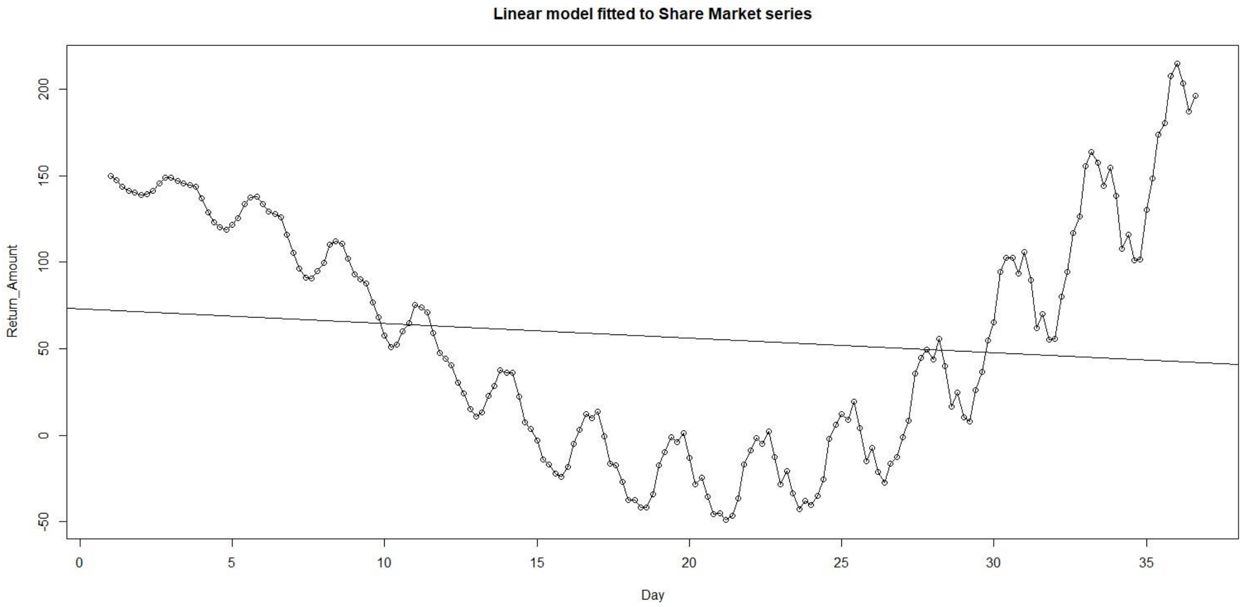

Figure 20

Inference: from the linear model plot, it is evident that our model fitting isn't suitable for share market series.

Residual analysis of linear model

```
# Residual Analysis of linearmodel
res.linearmodel = rstudent(linearmodel)
par(mfrow=c(3,3))
plot(y = res.linearmodel, x = as.vector(time(ShareMarketTS)),xlab = 'Day', ylab='Standardized Residuals',type='l',main = "Standardised
hist(res.linearmodel, main = "Histogram of standardised residuals from linear model.")
qqnorm(y=res.linearmodel, main = "QQ plot of standardised residuals from linear model.")
qqline(y=res.linearmodel, col = 2, lwd = 1, lty = 2)
shapiro.test(res.linearmodel)
acf(res.linearmodel, main = "ACF of standardized residuals from linear model.")
pacf(res.linearmodel, main = "PACF of standardized residuals from linear model.")
graphics.off()
```

Figure 21

Output

```
> shapiro.test(res.linearmodel)

        Shapiro-Wilk normality test

data:  res.linearmodel
W = 0.9525, p-value = 1.022e-05
```

Figure 22

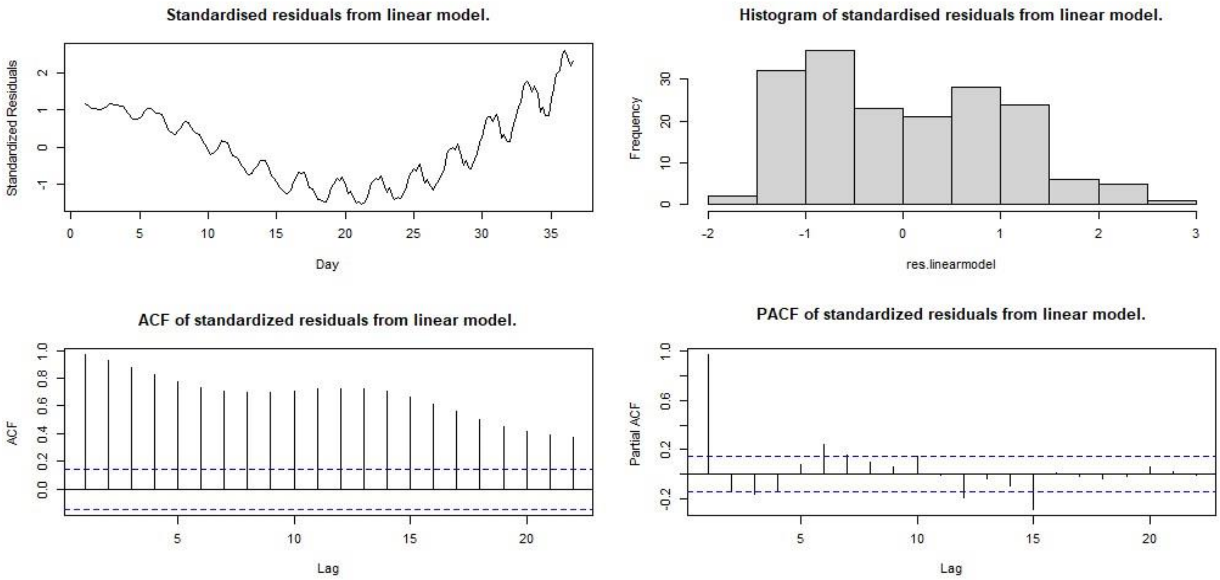

Figure 23

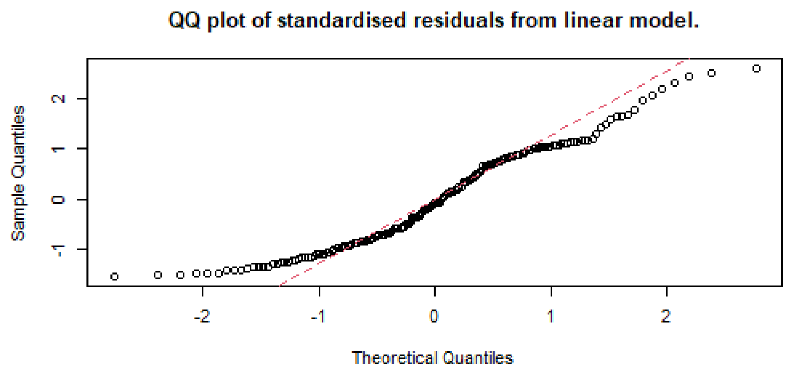

Figure 24

Inference: according to the Shapiro wilk test we want a p-value to be greater than 0.05 to not reject the null hypothesis which says that the quantiles are normally distributed but here we have a value of 1.022e-05 which is very close to zero. Hence, we reject the null hypothesis and conclude that residuals of linear model are significantly non-normally distributed. If we look at the histogram plot of standardised residual of linear model, it isn't forming a symmetric normal distribution curve. Q-Q plot also doesn't fit all the data points on the reference line, we have many outliers in the plot. In the ACF plot of standardised residuals, all the bars are significantly outside the confidence limits, whereas we want all of them to be within the confidence limits. PACF plot also has some significant lags in the standardised residuals, it doesn’t show randomness.

2. Quadratic model

Fitting the quadratic model

```
# Fitting the quadratic model
t = time(ShareMarketTS)
t2 = t^2
quadmodel = lm(ShareMarketTS~ t + t2)
summary(quadmodel)
```

Figure 25

Output

```
> summary(quadmodel)

Call:
lm(formula = ShareMarketTS ~ t + t2)

Residuals:
    Min      1Q  Median      3Q     Max 
-59.258 -19.321   0.571  19.992  53.165 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 233.98184    6.50637   35.96   <2e-16 ***
t           -25.40069    0.79605  -31.91   <2e-16 ***
t2            0.65312    0.02056   31.77   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 26.27 on 176 degrees of freedom
Multiple R-squared:  0.8539,	Adjusted R-squared:  0.8523 
F-statistic: 514.4 on 2 and 176 DF,  p-value: < 2.2e-16
```

Figure 26

Inference: the coefficient of intercept for quadratic model has come out to be 233.98184 when both t and t^2 are zero and coefficient of linear term t is -25.40069 and coefficient of quadratic term t2 is 0.65312, these coefficients indicate a concave upward trend, all the coefficients are statistically significant for this model. F-statistics has a very small p-value <2.2e-16 which is less than 0.05 suggesting that it is statistically significant. Adjusted R-squared for this model is 0.8523 which is high approximately 85%, suggesting that this quadratic model would be a good fit for share market series.

```
fitted.quadmodel <- fitted(quadmodel)
plot(ts(fitted.quadmodel), ylim = c(min(c(fitted(quadmodel), as.vector(ShareMarketTS))), max(c(fitted(quadmodel),
    as.vector(ShareMarketTS)))), ylab='y' , main = "Fitted quadratic curve to Share Market series", type="l",lty=2,col="red")
lines(as.vector(ShareMarketTS),type="o")
```

Figure 27

Output

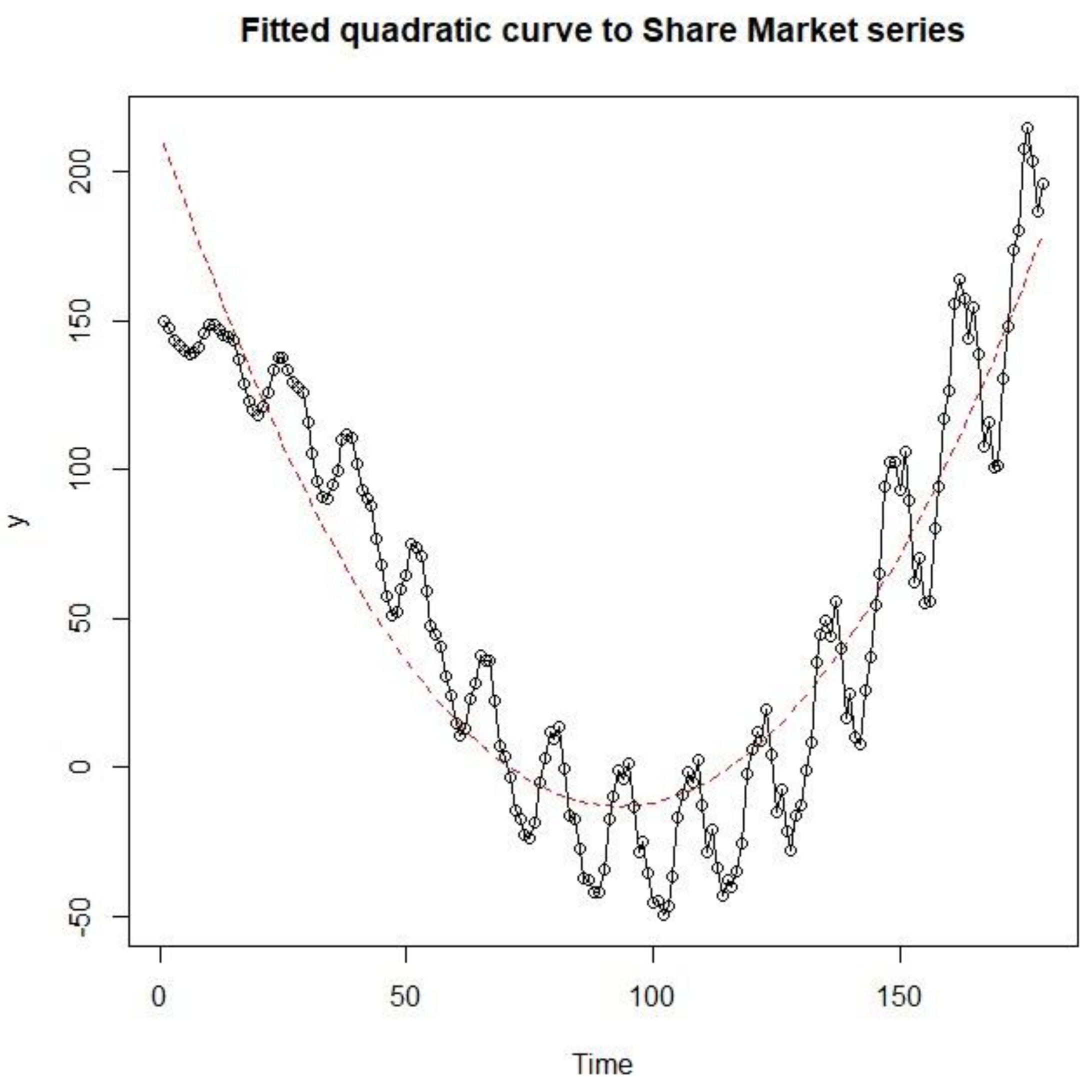

Figure 28

Inference: from the above plot, quadratic model is fitting almost perfectly into our share market series. We can say that quadratic model might be a good fit for our series.

Residual analysis of quadratic model

```
plot(y = res.quadmodel, x = as.vector(time(ShareMarketTS)),xlab = 'Time',
     ylab='Standardized Residuals',type='l',main = "Standardised residuals from quadratic model.")
hist(res.quadmodel,xlab='Standardized Residuals', main = "Histogram of standardised residuals from quadratic model.")
qqnorm(y=res.quadmodel, main = "QQ plot of standardised residuals from quadratic model.")
qqline(y=res.quadmodel, col = 2, lwd = 1, lty = 2)
shapiro.test(res.quadmodel)
par(mfrow=c(2,1))
acf(res.quadmodel, main = "ACF of standardized residuals from quadratic model.")
pacf(res.quadmodel, main = "PACF of standardized residuals from quadratic model.")
```

Figure 29

Output

```
> shapiro.test(res.quadmodel)

        Shapiro-Wilk normality test

data:  res.quadmodel
W = 0.98397, p-value = 0.03799
```

Figure 30

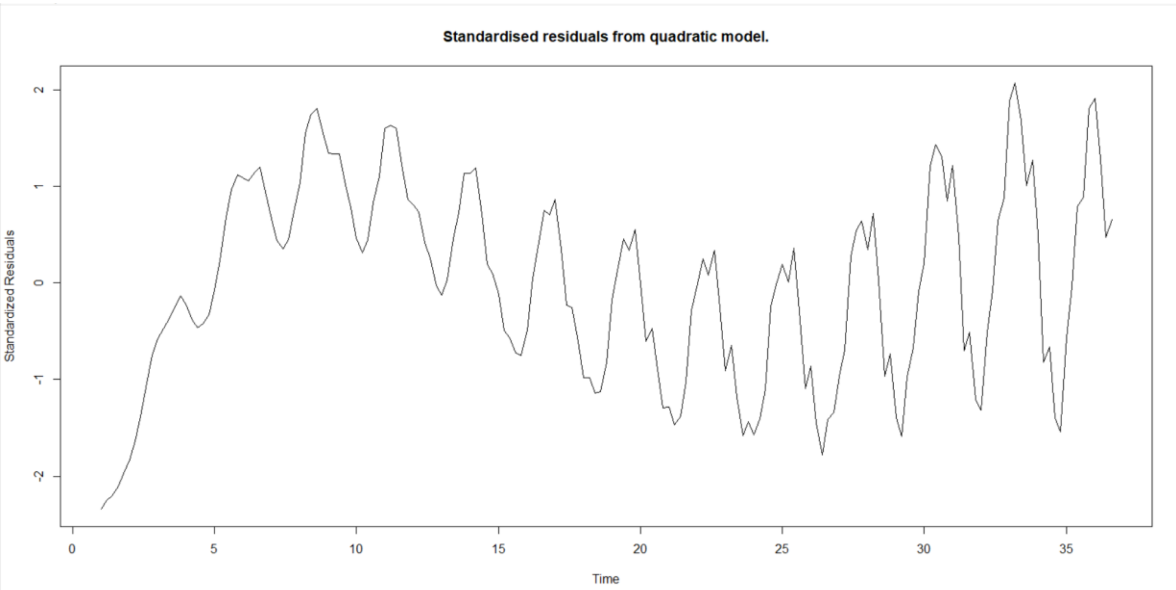

Figure 31

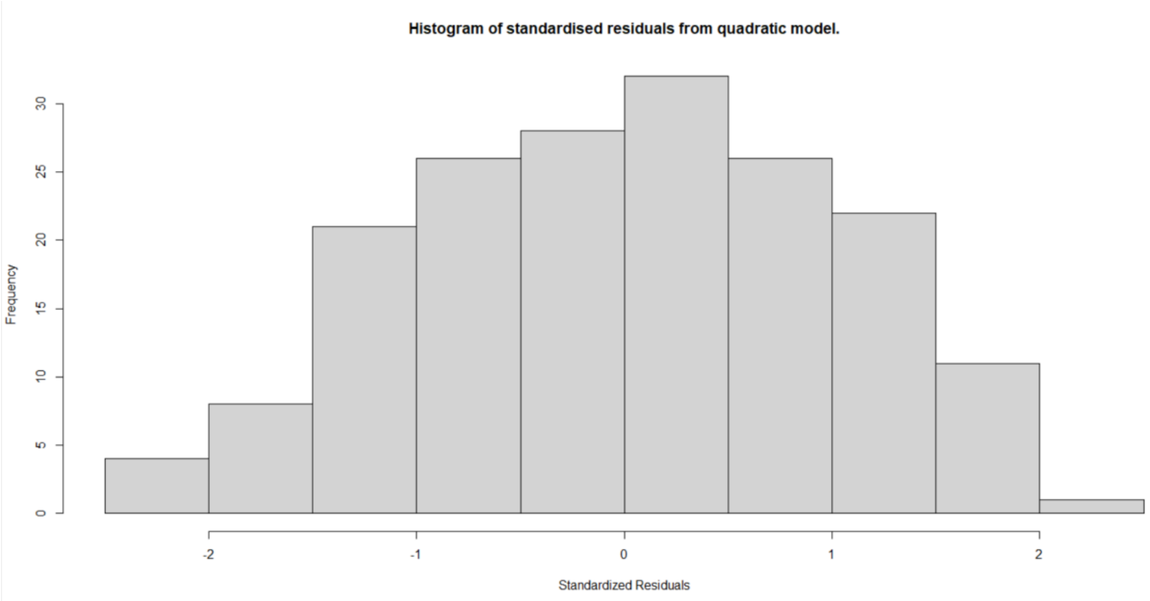

Figure 32

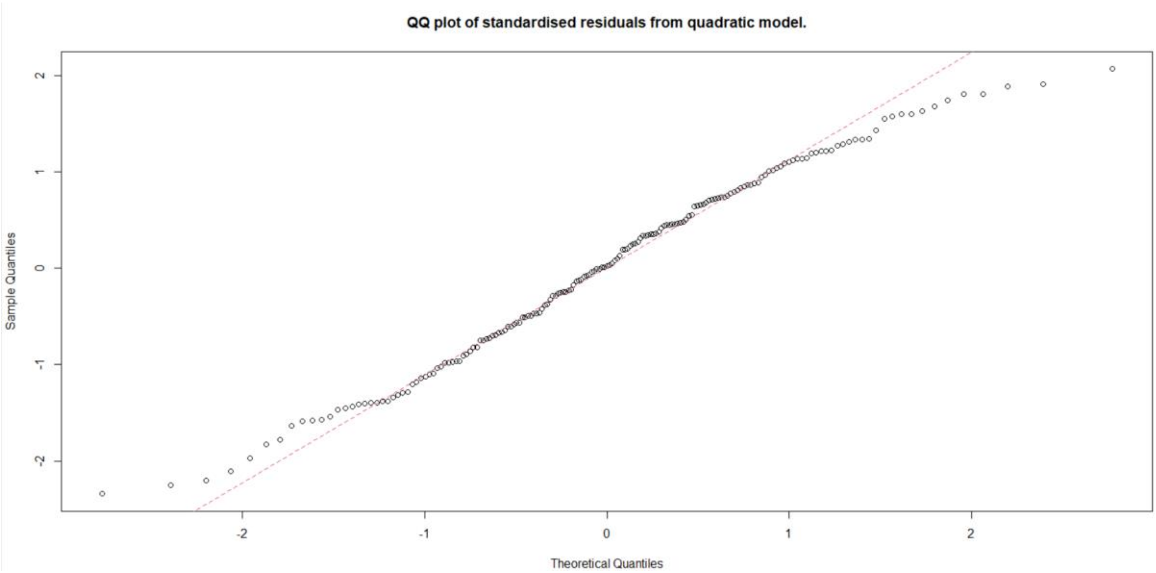

Figure 33

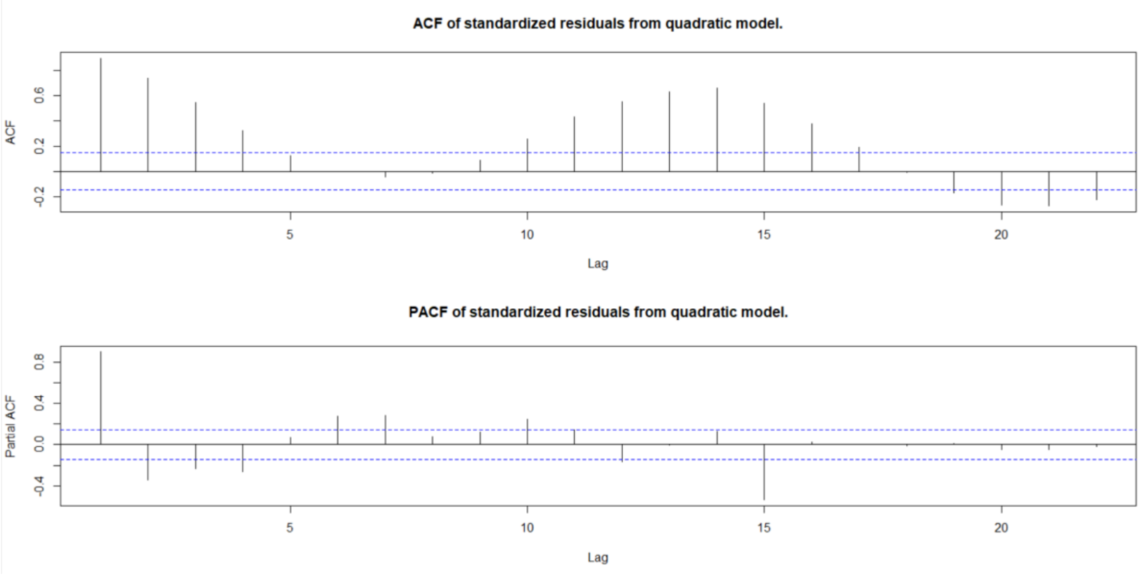

Figure 34

Inference: although from Shapiro wilk test of residuals from quadratic model we get a p-value of 0.03799 which is less than 0.05 hence, we reject the null hypothesis saying that distribution isn't normal but if we look at the histogram plot of the residuals, we see a normal curve forming and the histogram is symmetric. Q-Q plot is not able to fit all the points in reference line. ACF plot has many significant lags and have a pattern forming suggesting some seasonality, which our quadratic model can't fit. PACF plot also has some significant lags, but it does show some randomness in the pattern.

3. Seasonal model

```r
# Fitting the Seasonal model
day.= season(ShareMarketTS)
dataFreq <- frequency(ShareMarketTS)
seasonalmodel=lm(ShareMarketTS~day.-1)
summary(seasonalmodel)
```

Figure 35

Output

```
> seasonalmodel=lm(ShareMarketTS~day.-1)
> summary(seasonalmodel)

Call:
lm(formula = ShareMarketTS ~ day. - 1)

Residuals:
     Min       1Q   Median       3Q      Max
-106.785  -60.423   -5.695   62.046  156.912

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
day.Season-1    57.70      11.52   5.008 1.34e-06 ***
day.Season-2    57.62      11.52   5.001 1.38e-06 ***
day.Season-3    57.83      11.52   5.020 1.27e-06 ***
day.Season-4    57.78      11.52   5.015 1.30e-06 ***
day.Season-5    54.21      11.68   4.639 6.85e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 69.13 on 174 degrees of freedom
Multiple R-squared:  0.4121,	Adjusted R-squared:  0.3952
F-statistic: 24.39 on 5 and 174 DF,  p-value: < 2.2e-16
```

Figure 36

Inference: all the coefficients of the seasonal model are statistically significant at 5% level, which says that the seasonal factor has a significant effect on the series. F-statistics as a p-value of <2.2e-16, which is less then 0.05, so it is statistically significant. Adjusted R-squared for seasonal model is 0.3952, which is low so maybe seasonal model might not be a very good fit for the series. This model is a moderate fit for our series.

```
fitted.seasonalmodel <- fitted(seasonalmodel)
plot(ts(fitted(seasonalmodel)), ylab='y', main = "Fitted seaonal model to Share Market series.",
     ylim = c(min(c(fitted(seasonalmodel), as.vector(ShareMarketTS))) ,
              max(c(fitted(seasonalmodel), as.vector(ShareMarketTS)))
     ), col = "red" )
lines(as.vector(ShareMarketTS),type="o")
```

Figure 37

Output

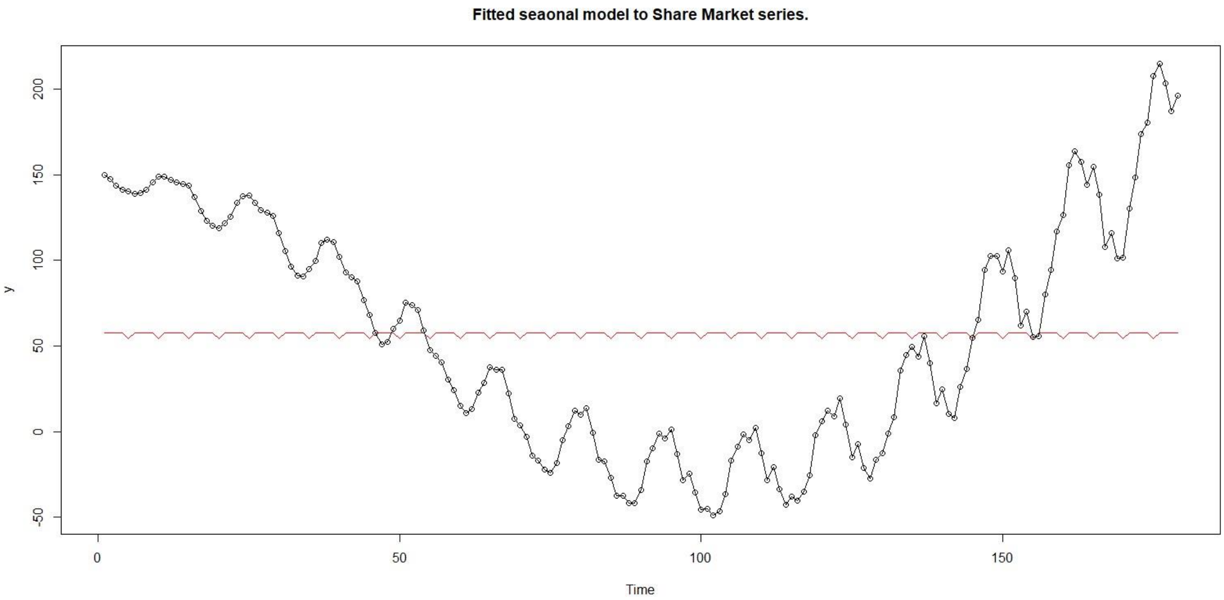

Figure 38

Inference: fitted seasonal model plot is coming out to be very inconsistent, fitted line isn't following the trend the series and is neither able to fit in the seasonality of the model.

Residual analysis of seasonal model

```
# Residual Analysis
res.seasonalmodel = rstudent(seasonalmodel)
plot(y = res.seasonalmodel, x = as.vector(time(ShareMarketTS)),xlab = 'Time',
     ylab='Standardized Residuals',type='l',main = "Standardised residuals from seasonal model.")
points(y=res.seasonalmodel,x=time(ShareMarketTS), pch=as.vector(season(ShareMarketTS)))
hist(res.seasonalmodel,xlab='Standardized Residuals', main = "Histogram of standardised residuals from seasonal model.")
qqnorm(y=res.seasonalmodel, main = "QQ plot of standardised residuals from seasonal model.")
qqline(y=res.seasonalmodel, col = 2, lwd = 1, lty = 2)
shapiro.test(res.seasonalmodel)
par(mfrow=c(2,1))
acf(res.seasonalmodel, main = "ACF of standardized residuals from seasonal model.")
pacf(res.seasonalmodel, main = "PACF of standardized residuals from seasonal model.")
```

Figure 39

Output

```
        Shapiro-Wilk normality test

data:  res.seasonalmodel
W = 0.94757, p-value = 3.631e-06
```

Figure 40

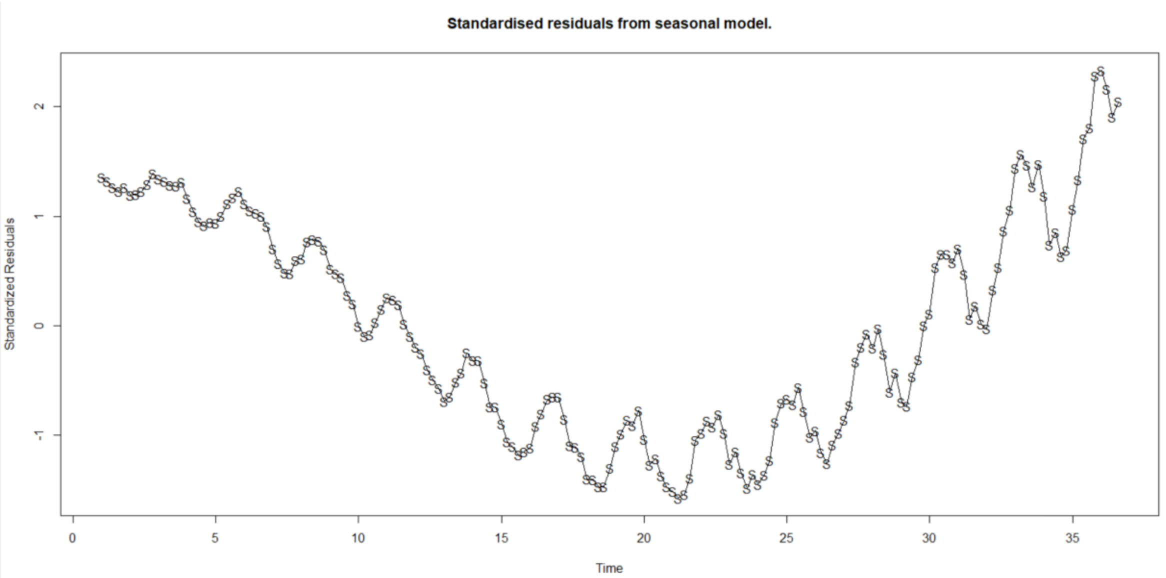

Figure 41

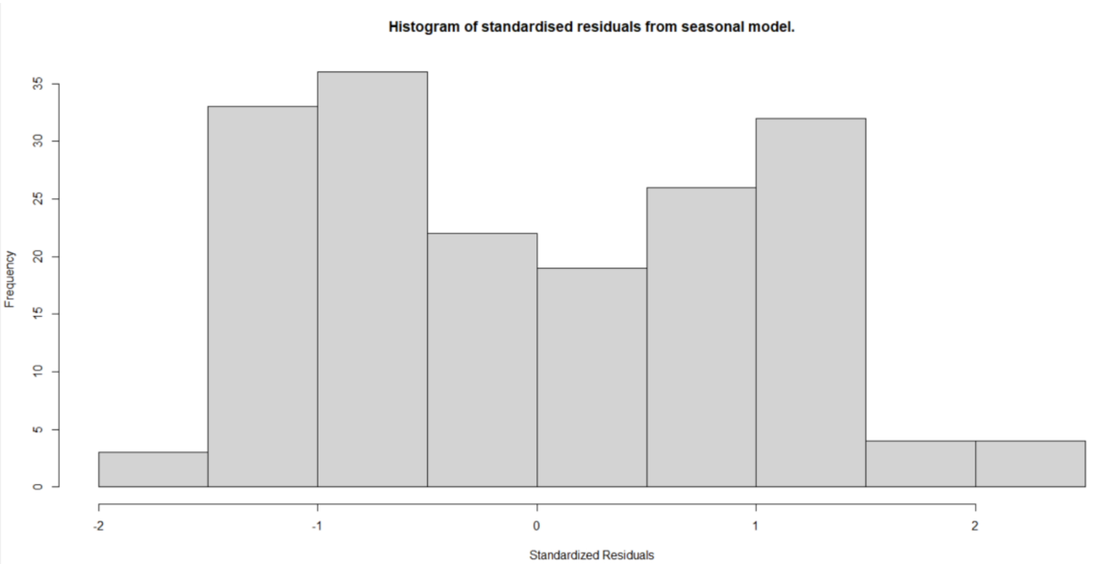

Figure 42

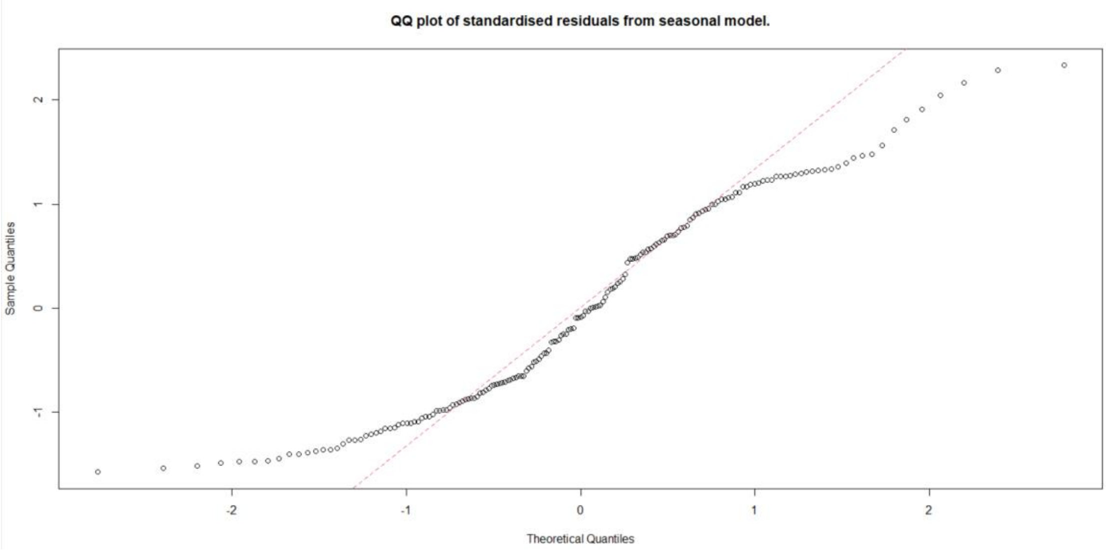

Figure 43

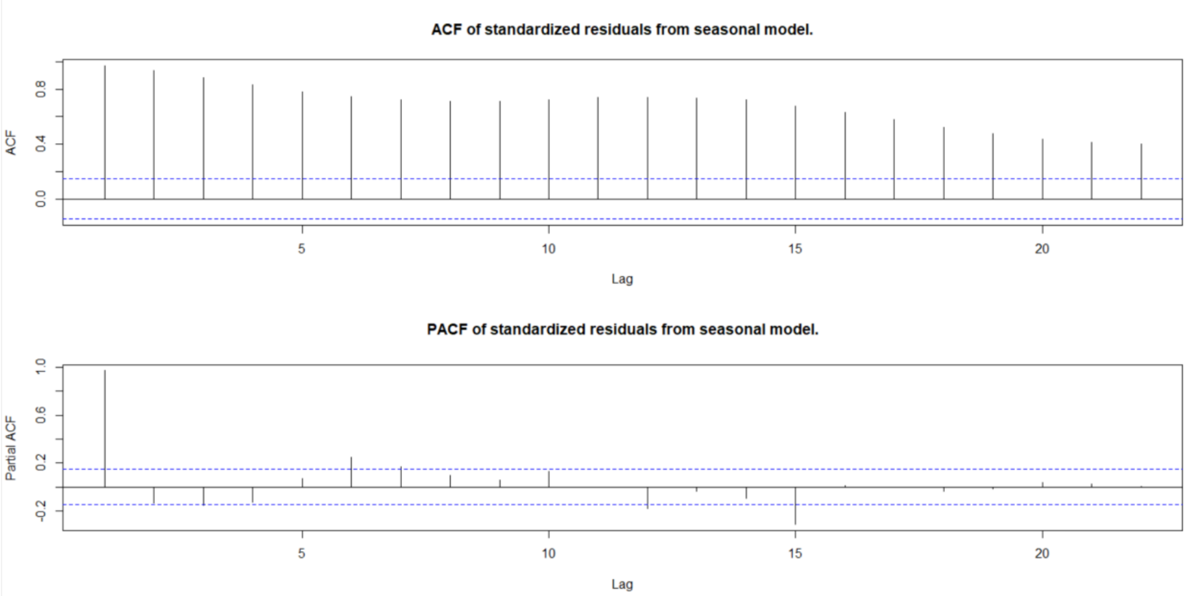

Figure 44

Inference: Shapiro wilk test has a p-value of 3.631e-06 which is smaller than 0.05 hence we reject the null hypothesis, seasonal model's residuals are not normally distributed. We can back this conclusion by looking at the histogram plot of the standardised residuals from seasonal model, the bars of the histogram aren't normally distributed and are asymmetrical. In Q-Q plot all the data points aren't falling on the reference line and there are many outliers. ACF plot has many significant lags i.e. bars being out of the confidence level. PACF plot also has some significant lags.

4. Quadratic plus seasonal model

```
# Fitting quadratic + seasonal model
day. <- season(ShareMarketTS)
t <- time(ShareMarketTS)
t2 <- t^2
modelqs <- lm(ShareMarketTS ~ day. + t + t2 -1)
summary(modelqs)
```

Figure 45

Output

```
> modelqs <- lm(ShareMarketTS ~ day. + t + t2 -1)
> summary(modelqs)

Call:
lm(formula = ShareMarketTS ~ day. + t + t2 - 1)

Residuals:
    Min      1Q  Median      3Q     Max 
-58.888 -19.624   0.437  19.776  53.412 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
day.Season-1 233.6141     7.6379   30.59   <2e-16 ***
day.Season-2 233.7541     7.6570   30.53   <2e-16 ***
day.Season-3 234.1375     7.6751   30.51   <2e-16 ***
day.Season-4 234.1987     7.6923   30.45   <2e-16 ***
day.Season-5 234.2983     7.7848   30.10   <2e-16 ***
t            -25.4025     0.8054  -31.54   <2e-16 ***
t2             0.6532     0.0208   31.40   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 26.58 on 172 degrees of freedom
Multiple R-squared:  0.9141,	Adjusted R-squared:  0.9106 
F-statistic: 261.5 on 7 and 172 DF,  p-value: < 2.2e-16
```

Figure 46

Inference: all the coefficients of the quadratic plus seasonal model are statistically significant at 5% level. F-statistics at 261.5 has a p-value of <2.2e-16, which is less than 0.05, this suggests that the model is statistically significant and suggests that that the combination of quadratic plus seasonal model improves the fit of the model. R-squared value of the model is 0.9106 which is approximately 91%, this value is high and very close to 1 so this model might be overfitting the series.

```
fitted.modelqs<- fitted(modelqs)
plot(ts(fitted(modelqs)), ylim = c(min(c(fitted(modelqs), as.vector(ShareMarketTS))),
                                   max(c(fitted(modelqs),as.vector(ShareMarketTS)))),
     ylab='y' , main = "Fitted seasonal plus quadratic curve to Share Market series", type="l",lty=2,col="red")
lines(as.vector(ShareMarketTS),type="o")
```

Figure 47

Output

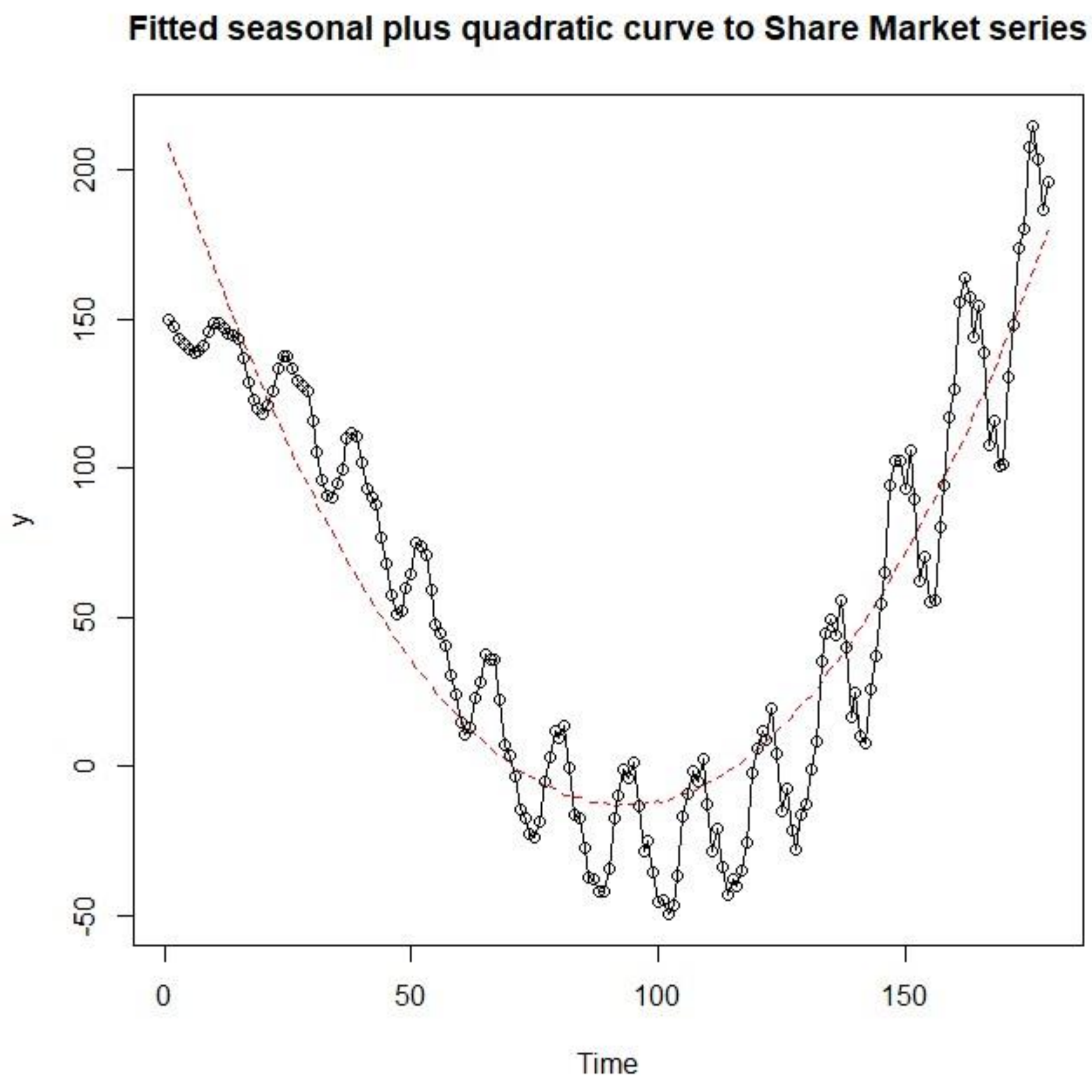

Figure 48

Inference: fitted quadratic plus seasonal model plot is relatively consistent as compared to other models before, it follows the trend of the series, but this model is still not able to completely cover the seasonality in the series.

Residual analysis of quadratic plus seasonal model

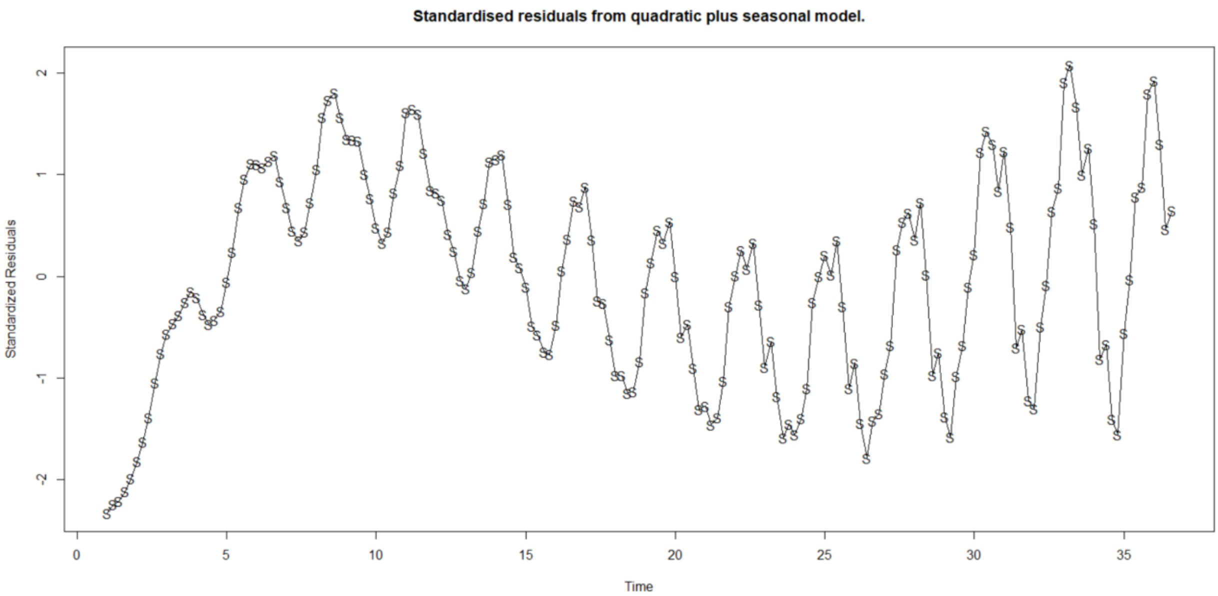

Figure 49

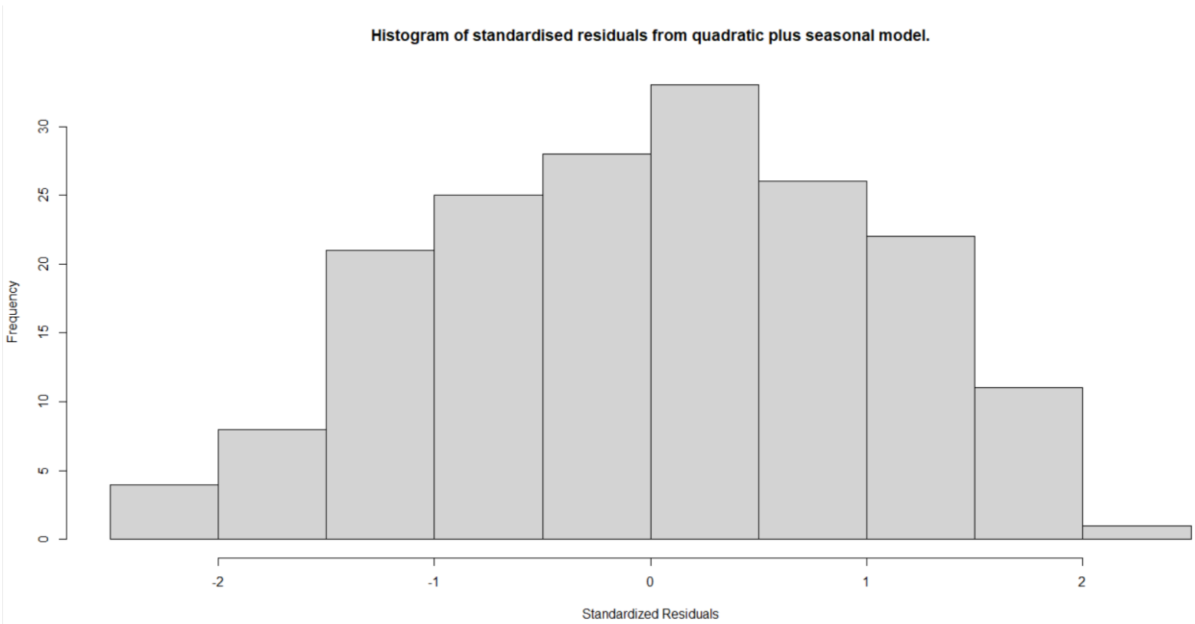

Figure 50

```
> shapiro.test(res.modelqs)

        Shapiro-Wilk normality test

data:  res.modelqs
W = 0.98409, p-value = 0.03944
```

Figure 51

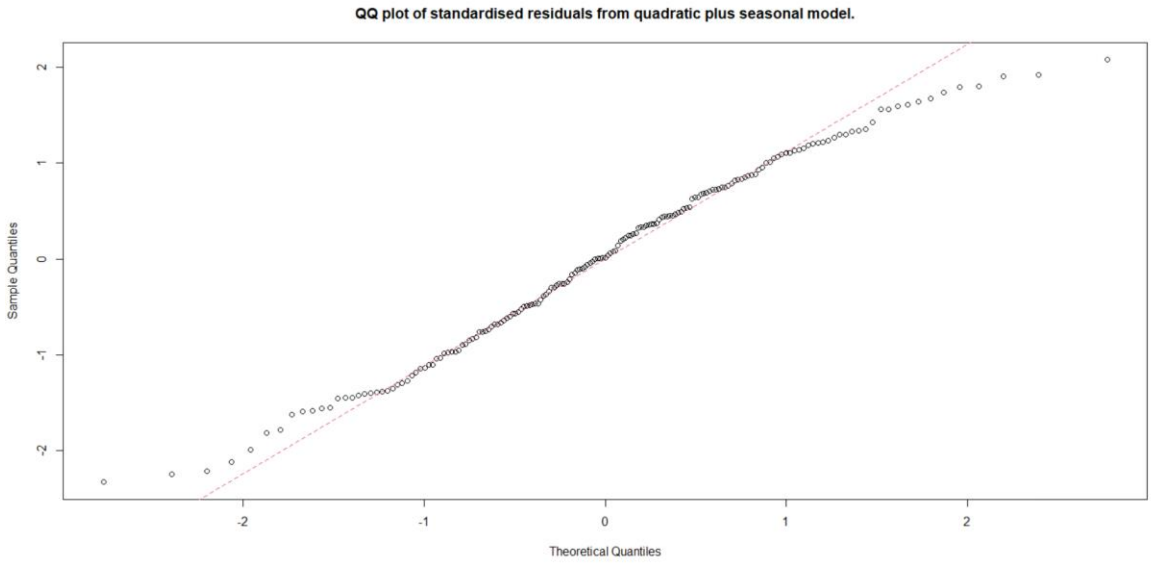

Figure 52

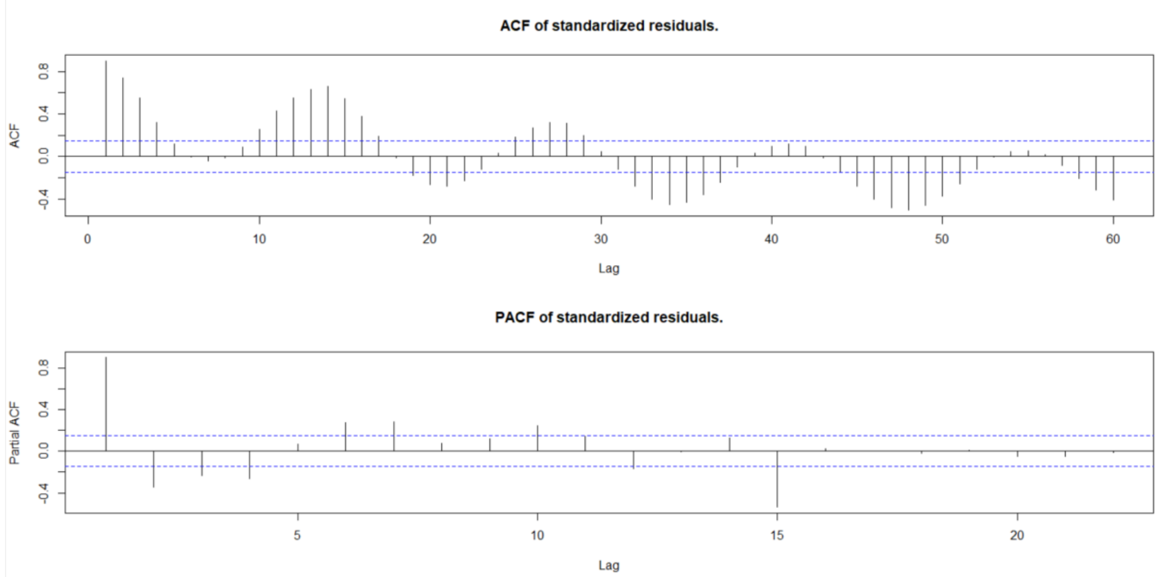

Figure 53

Inference: Shapiro wilk test has a p-value of 0.03944 which is smaller than 0.05 hence we reject the null hypothesis, quadratic plus seasonal model's residuals is not normally distributed. However, by looking at the histogram plot of the standardised residuals from quadratic plus seasonal model, the bars of the histogram are normally distributed and are almost symmetrical. In Q-Q plot all the data points aren't falling on the reference line and there are some outliers. ACF plot has many significant lags i.e. bars being out of the confidence level and does have a pattern showing seasonal behaviour. PACF plot also has some significant lags.

5. Cosine/harmonic model

```
# Fitting the cosine/harmonic model
har.=harmonic(ShareMarketTS, 1)
data <- data.frame(ShareMarketTS,har.)
harmonicmodel = lm(ShareMarketTS ~ cos.2.pi.t. + sin.2.pi.t. , data = data)
summary(harmonicmodel)
```

Figure 54

Output

```
> summary(harmonicmodel)

Call:
lm(formula = ShareMarketTS ~ cos.2.pi.t. + sin.2.pi.t., data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-107.278  -59.716   -6.408   61.674  158.081 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  57.0348     5.1379  11.101   <2e-16 ***
cos.2.pi.t.  -0.5054     7.2496  -0.070    0.945    
sin.2.pi.t.   1.2955     7.2826   0.178    0.859    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 68.74 on 176 degrees of freedom
Multiple R-squared:  0.0002069,	Adjusted R-squared:  -0.01115 
F-statistic: 0.01821 on 2 and 176 DF,  p-value: 0.982
```

Figure 55

Inference: coefficients of harmonic model, cos.2.pi.t and sin.2.pi.t are -0.5054 and 1.2955 which are both statistically insignificant as their p-value being 0.945 and 0.859 respectively are greater than 0.05. F-statistics on 0.01821 is 0.982 which is again higher than 0.05. R-squared value of this model is very low, -0.01115, suggesting that this model will be a bad fit for our share market series.

```
fitted.harmonicmodel <- fitted(harmonicmodel)
plot(ts(fitted(harmonicmodel)), ylim = c(min(c(fitted.harmonicmodel),as.vector(ShareMarketTS)),
                                          max(c(fitted.harmonicmodel),as.vector(ShareMarketTS))),
     ylab='y' , main = "Fitted cosine curve on Share Market series", type="l",lty=2,col="red")
lines(as.vector(ShareMarketTS),type="o")
```

Figure 56

Output

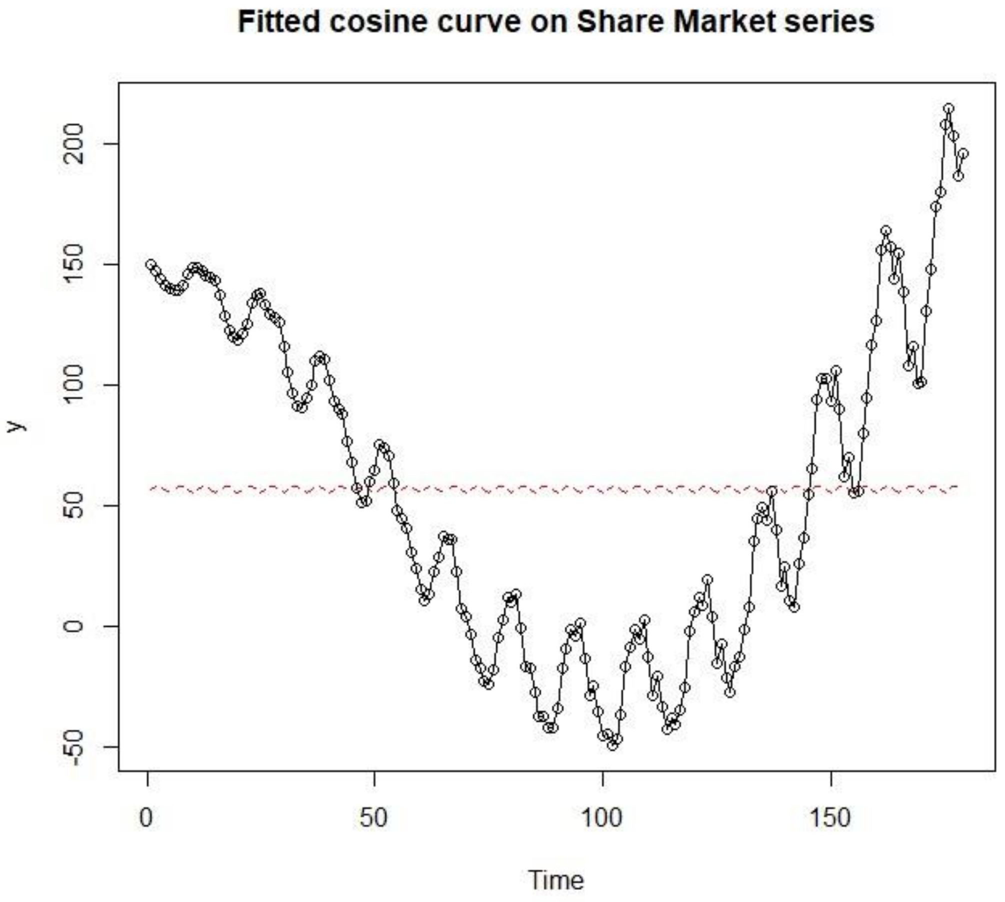

Figure 57

Inference: harmonic model fitted plot is very inconsistent, this model doesn't capture the trend at all of the share market series.

Residual analysis of harmonic model

```
# Residual Analysis
res.harmonicmodel = rstudent(harmonicmodel)
plot(y = res.harmonicmodel, x = as.vector(time(ShareMarketTS)),xlab = 'Time', ylab='Standardized Residuals',type='l'
     ,main = "Standardised residuals from harmonic model.")
points(y=res.harmonicmodel,x=time(ShareMarketTS.ts), pch=as.vector(season(ShareMarketTS)))
hist(res.harmonicmodel,xlab='Standardized Residuals', main = "Histogram of standardised residuals from harmonic model.")
qqnorm(y=res.harmonicmodel, main = "QQ plot of standardised residuals from harmonic model.")
qqline(y=res.harmonicmodel, col = 2, lwd = 1, lty = 2)
shapiro.test(res.harmonicmodel)
par(mfrow=c(2,1))
acf(res.harmonicmodel, main = "ACF of standardized residuals from harmonic model.")
pacf(res.harmonicmodel, main = "PACF of standardized residuals from harmonic model.")
```

Figure 58

Output

```
> shapiro.test(res.harmonicmodel)

        Shapiro-Wilk normality test

data:  res.harmonicmodel
W = 0.94756, p-value = 3.628e-06
```

Figure 59

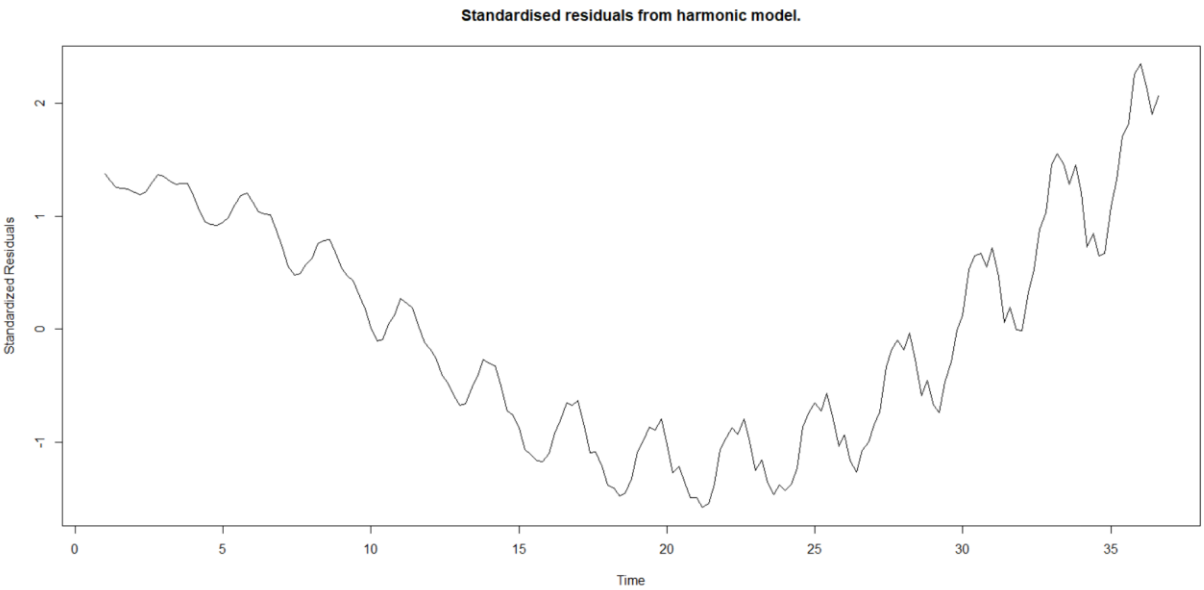

Figure 60

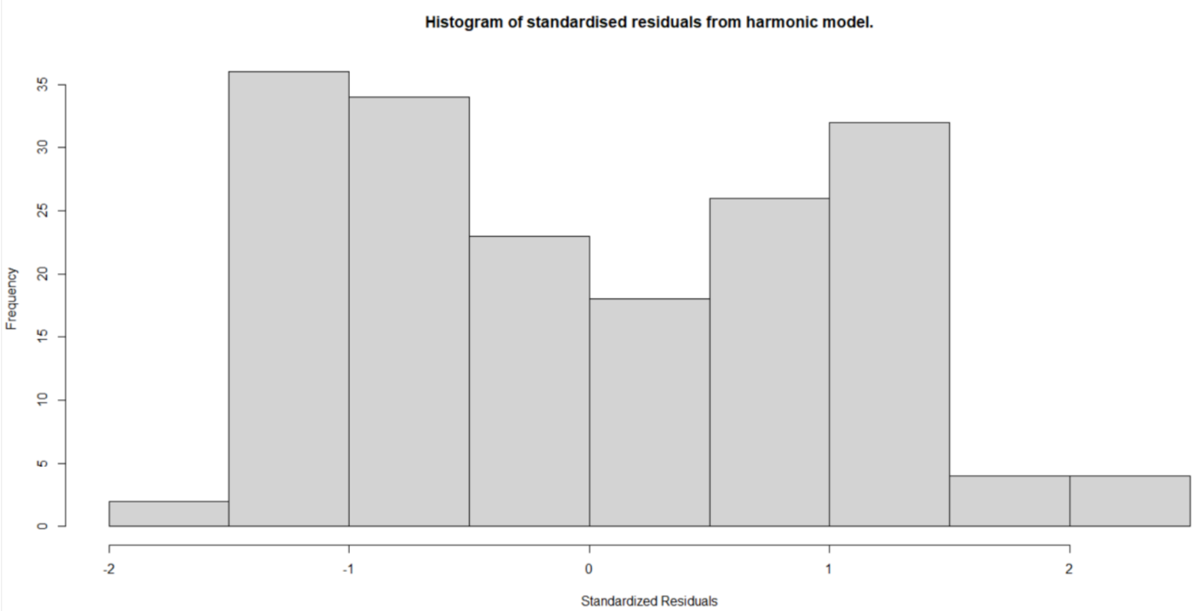

Figure 61

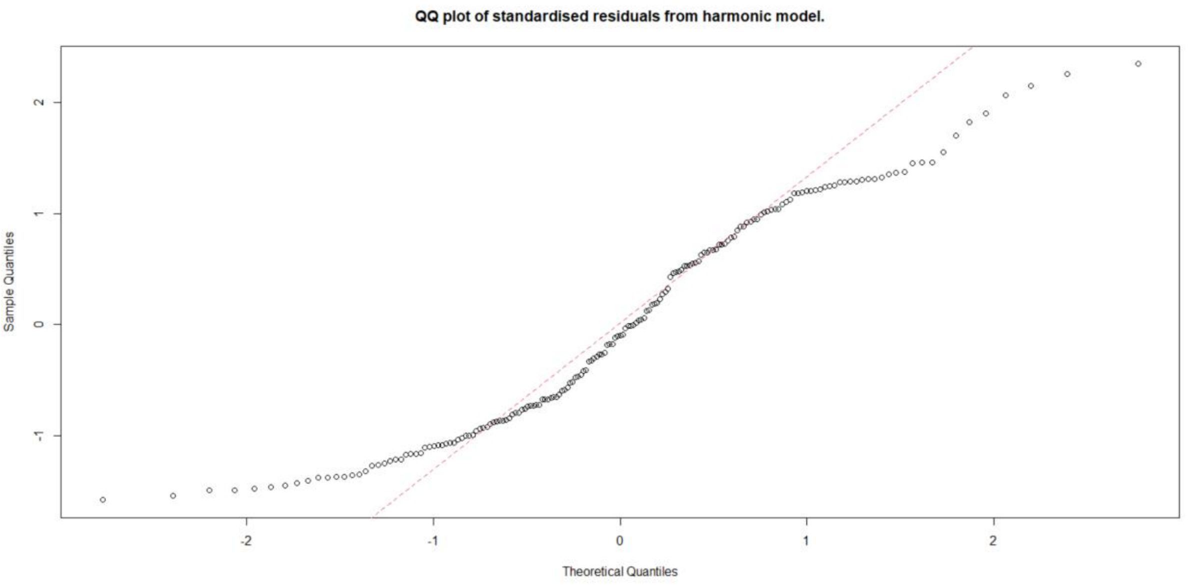

Figure 62

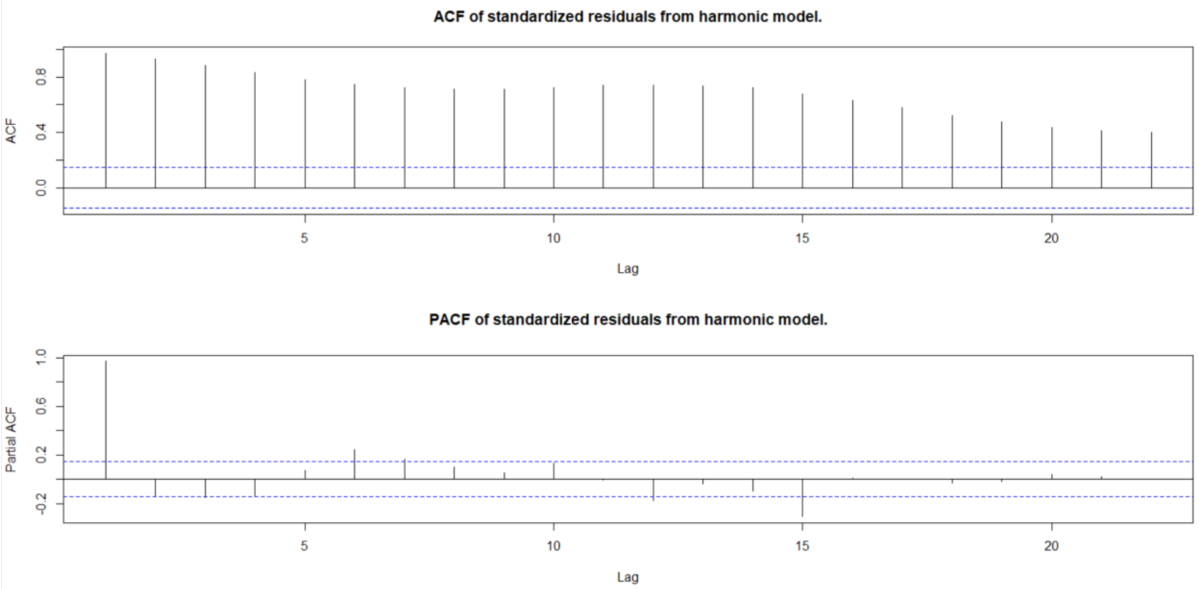

Figure 63

Inference: Shapiro wilk test's p-value is 3.628e-06 which is smaller than 0.05 hence we reject the null hypothesis and say that harmonic model's residuals aren't normally distributed. Histogram of harmonic model isn't symmetric and doesn't form a normal curve. Q-Q plot of harmonic model can't fit all the data points on the reference line. ACF plot has a wave pattern and there are many significant lags, whereas in the PACF plot there is some significant lag and shows randomness.

Conclusion from residual analysis

After fitting all the models i.e. linear, quadratic, seasonal, quadratic plus seasonal and harmonic model. Linear model, seasonal and harmonic models are very inconsistent with our share market trader investment portfolio series. The quadratic plus seasonal model is good fit for our series but it has an R-squared value approximately of 91% which is high and close to 1 or 100% so the model might be overfitting our series, although the fit captures the trend of the series, but the fit doesn't capture the seasonality in the series. Whereas with the quadratic model fitting we get a R-squared value of approximately 85% which shows that is a good fit for the series, the quadratic model also adheres with the normal distribution and in comparison, to both the models there isn't much difference between the two models. Hence, we take the quadratic model forward because it the best fit model out of all and do the next five trading days predictions using the quadratic model.

Forecasting

```
# FORECASTING FROM THE BEST FIT MODEL
# Next five trading days forecast
h <- 5
lastTimePoint <- t[length(t)]
aheadTimes <- data.frame(t = seq(lastTimePoint+(1/dataFreq), lastTimePoint+h*(1/dataFreq), 1/dataFreq),
                         t2 =  seq(lastTimePoint+(1/dataFreq), lastTimePoint+h*(1/dataFreq), 1/dataFreq)^2)

frcquadmodel <- predict(quadmodel, newdata = aheadTimes, interval = "prediction")

plot(ShareMarketTS, xlim= c(t[1],aheadTimes$t[nrow(aheadTimes)]), ylim = c(-150,300), ylab = "Share Market series", xlab="days",
     main = "Forecasts from the quadratic model fitted to the  Share Market series.")
lines(ts(fitted.quadmodel,start = t[1],frequency = dataFreq))
lines(ts(as.vector(frcquadmodel[,3]), start = aheadTimes$t[1],frequency = dataFreq), col="dark green", type="l")
lines(ts(as.vector(frcquadmodel[,1]), start = aheadTimes$t[1],frequency = dataFreq), col="maroon", type="l")
lines(ts(as.vector(frcquadmodel[,2]), start = aheadTimes$t[1],frequency = dataFreq), col="dark green", type="l")
legend("topleft", lty=1, pch=1, col=c("black","dark green","maroon"),
       c("Data","5% forecast limits", "Forecasts"))
```

Figure 64

Output

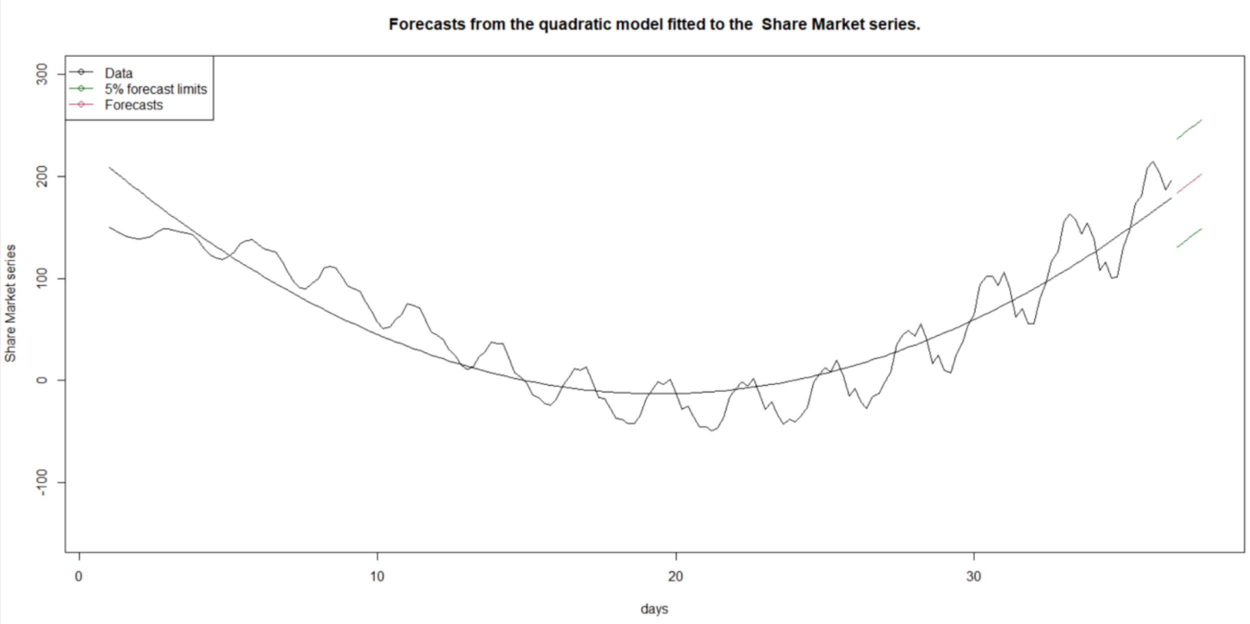

Figure 65

Summary

In this report we have used the time series analysis regression trend analysis for model building. All the statistical computation was done using R studio using TSA library package. We analysed the dataset of share market traders' investment portfolio, various model fitting, and model diagnosis was done. After interpreting all the models, we came up with the best fit model for the series and forecasting was done using that model for predicting the next five trading days.

References

Dataset used
https://rmit.instructure.com/courses/124176/files/36189467/download?download_frd=1

Reading materials used

Module 1 presentation:

https://rmit.instructure.com/courses/124176/files/37065155/download?download_frd=1

Module 1 notes:

https://rmit.instructure.com/courses/124176/files/36179161/download?download_frd=1

Module 2 presentation:

https://rmit.instructure.com/courses/124176/files/37079754/download?download_frd=1

Module 2 notes:

https://rmit.instructure.com/courses/124176/files/36178912/download?download_frd=1

R code references:

https://rmit.instructure.com/courses/124176/files/37079799/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37172225/download?download_frd=1